

EDA report on Student Performance Dataset

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Abstract—This study investigates students’ performance based on various factors including gender, race/ethnicity, and test preparation course completion. The purpose of this study is to analyze and compare academic outcomes using exploratory data analysis (EDA). The dataset contains both categorical and numerical variables, which are visualized using different graphs and plots to identify patterns, relationships, and trends among the features. These visualizations are then analyzed and compared to draw meaningful conclusions. The analysis reveals that students who complete the test preparation course generally score higher than those who do not. Additionally, female students tend to achieve higher writing scores compared to male students. Overall, the findings indicate that both demographic and preparatory factors play a significant role in influencing student academic performance.

I. INTRODUCTION

Student academic performance is influenced by a combination of cognitive, social, and environmental factors. Understanding these influences is essential for improving teaching strategies and educational outcomes. In recent years, data-driven approaches have become increasingly important in analyzing student performance and identifying key factors that contribute to academic success. The dataset used in this study contains exam scores in mathematics, reading, and writing, along with demographic attributes such as gender, race/ethnicity, parental level of education, lunch type, and test preparation course completion. These variables allow for a comprehensive analysis of both academic results and background influences. This study focuses on performing exploratory data analysis (EDA) to uncover patterns, trends, and relationships within the dataset. It aims to determine how different factors impact student performance and to identify any anomalies or outliers present in the data. The insights gained from this analysis can help educators and policymakers develop strategies to improve student learning outcomes and address performance gaps among different groups.

II. METHODOLOGY

A. Research Design

This study follows a quantitative research design using exploratory data analysis (EDA) [2]. The analysis is performed on a publicly available dataset containing student exam scores and demographic attributes. Statistical summaries and visualization techniques are used to explore relationships between variables and identify meaningful patterns.

B. Dataset Description

The dataset used in this study is the “Students Performance in Courses” dataset obtained from Kaggle [1]. It consists of 1000 student records with 8 attributes, including:

- Gender
- Race/ethnicity
- Parental level of education
- Lunch type
- Test preparation course
- Math score
- Reading score
- Writing score

The dataset includes both categorical and numerical variables.

C. Data Collection

The data was obtained from a publicly available Kaggle dataset. No primary data collection was performed. There were no missing data in this dataset. The dataset is structured and contains both categorical and numerical features related to student performance.

III. FINDINGS

The exploratory data analysis revealed several meaningful patterns in student performance.

As shown in Fig. 1 and Fig. 2, Students who completed the test preparation course scored significantly higher in mathematics compared to those who did not complete the course. This suggests that structured preparation has a positive impact on academic outcomes.

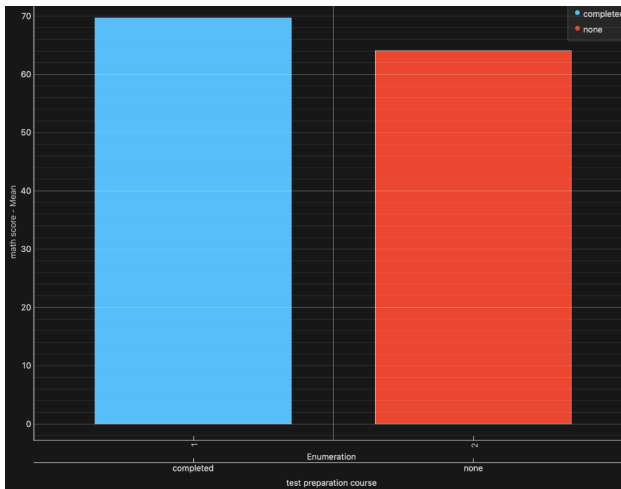


Fig. 1. Average math score by test preparation course completed

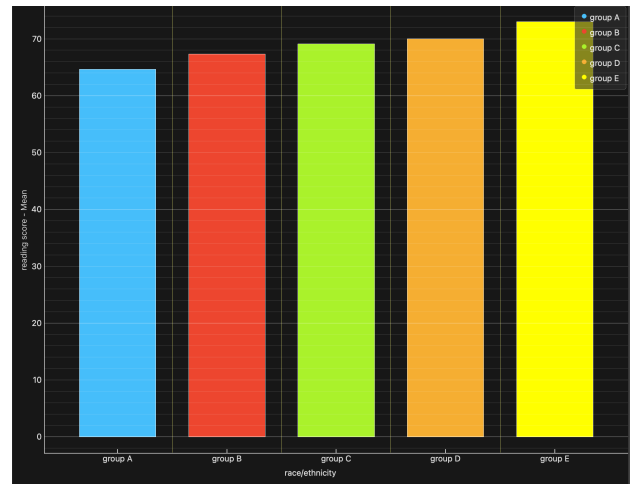


Fig. 3. Average reading score by race/ethnicity

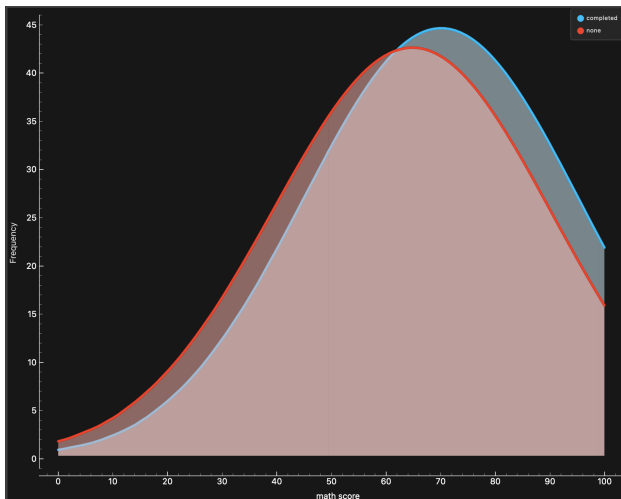


Fig. 2. Average math score by test preparation course(Density plot)

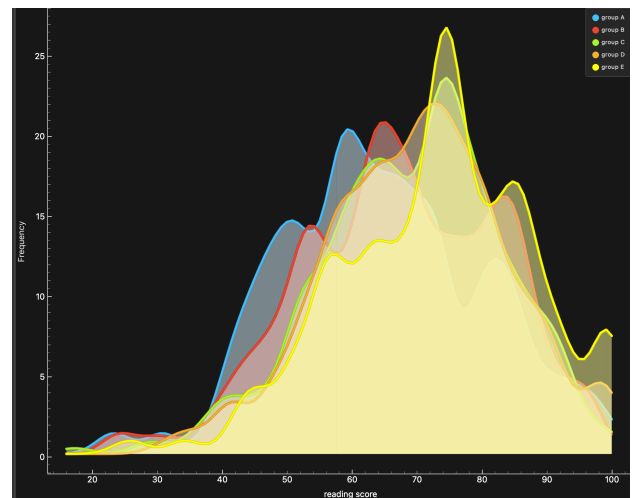


Fig. 4. Average reading score by race/ethnicity(Density plot)

As shown in Fig. 3 and Fig. 4, The average reading score varies across different race/ethnicity groups. Students in group E have the highest average reading score, followed by groups D and C, while group A has the lowest. This suggests a noticeable variation in academic performance across different groups. In Fig. 4, the density plot is shown at smoothing 9 to better understand the pattern.

As shown in Fig. 5 and Fig. 6, Gender-based analysis showed that female students generally have better in writing scores, while male students showed slightly decreased writing scores. This indicates a variation in subject-wise performance across genders.

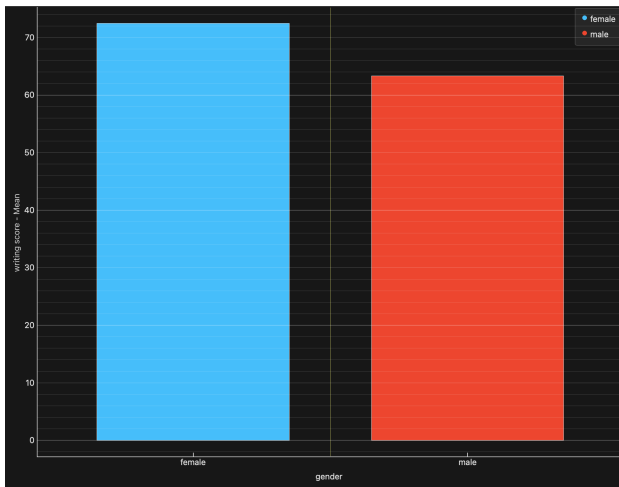


Fig. 5. Average writing score by gender(Bar plot)

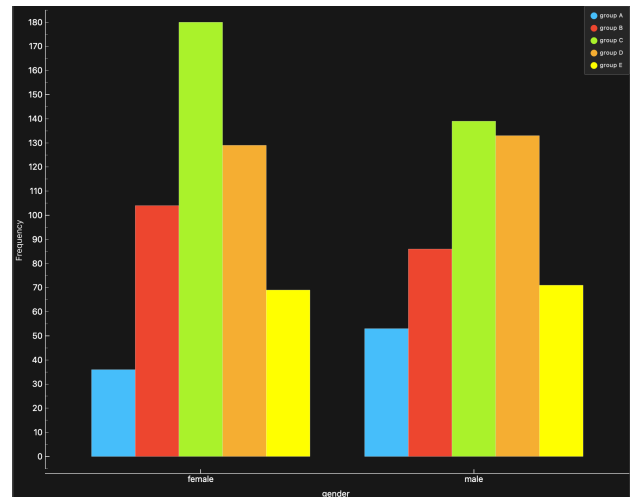


Fig. 7. Average number of female and male in all race/ethnicity

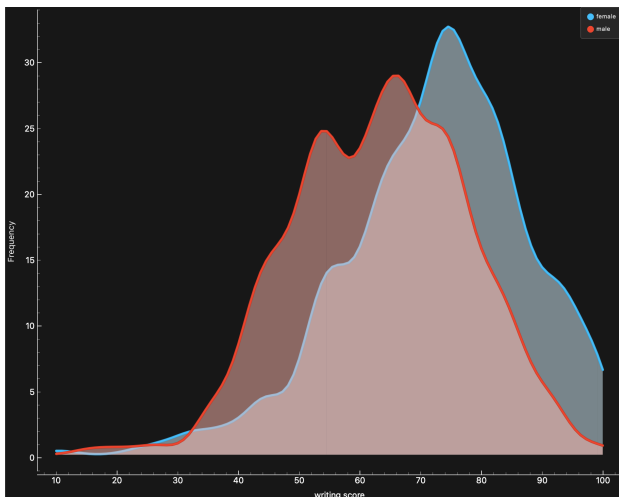


Fig. 6. Average number of writing score by gender(Density plot)

As shown in Fig. 7, The gender-wise histogram shows that female students have a higher concentration in mid-level score groups, especially Group C and Group D, compared to males. Male students are more distributed toward lower and mid categories, indicating slightly lower overall performance in comparison. This suggests a variation in academic achievement patterns across genders.

As shown in Fig. 8, This histogram, representing completion status, shows that the majority of students fall under the “none” category for both males and females, while a comparatively smaller number have completed the task. Females show slightly higher completion counts than males. This indicates generally low completion rates across both genders, with only minor gender differences.

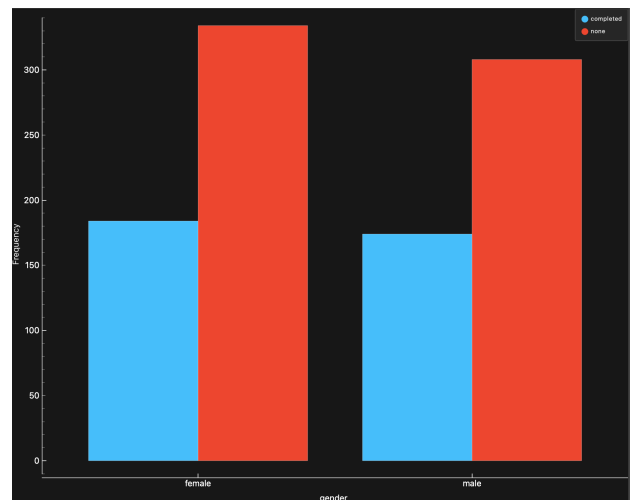


Fig. 8. Average number of females and male taking test preparation course

As shown in Fig. 9, scatter plot shows

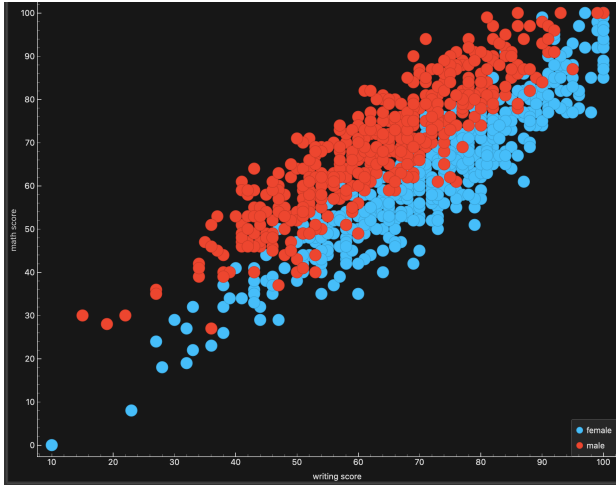


Fig. 9. Writing score and math score by attribute gender(Scatter plot)

As shown in Fig. 10 , scatter plot shows this ... last

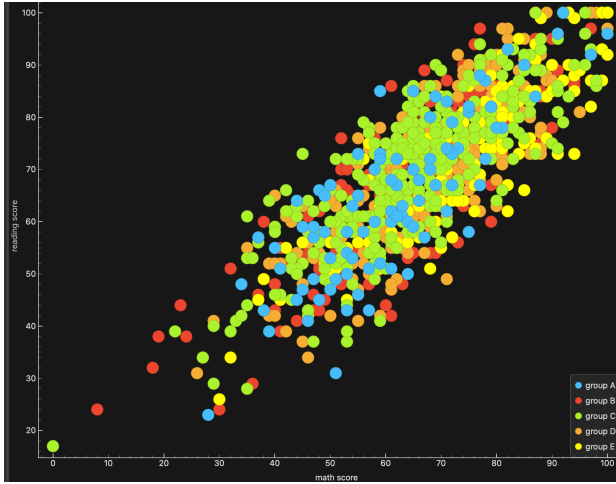


Fig. 10. Reading score and maths score by attribute race/ethnicity (Scatter plot)

IV. DISCUSSION

The findings from the exploratory data analysis are summarized and compared in Table I. This table highlights the relationship between different factors and student performance, making it easier to interpret overall trends.

TABLE I
SUMMARY OF KEY FINDINGS FROM EDA

Factor	Observation	Interpretation / Impact
Test Preparation Course	Students who completed the course scored higher in all subjects	Strong positive impact on academic performance, especially in math scores
Gender	Females performed better in writing; males slightly lower overall in writing	Indicates subject-wise gender variation in performance
Race/Ethnicity	Group E had highest reading scores; Group A lowest	Shows variation in performance across demographic groups
Lunch Type (if used in analysis)	Students with standard lunch performed better than free/reduced group	Suggests socio-economic influence on performance
Score Relationships	Strong positive correlation between math, reading, and writing scores	Students performing well in one subject tend to perform well in others
Test Completion Rate	Majority of students did not complete preparation course	Low participation may limit overall performance improvement

Overall Interpretation:

The analysis indicates that academic performance is influenced by both demographic and behavioral factors. Among all variables, completion of the test preparation course shows the most consistent positive effect across subjects. Gender and race/ethnicity show moderate variations, while subject scores are strongly correlated with each other. This suggests that improving study preparation and engagement can significantly enhance overall student performance.

V. CONCLUSION

This study performed an exploratory data analysis (EDA) on the Student Performance dataset to understand how demographic and behavioral factors influence academic outcomes[3]. The analysis revealed that students who completed the test preparation course consistently achieved higher scores across all subjects, highlighting the importance of structured academic preparation.

Gender-based analysis showed that female students performed slightly better in writing, while differences in mathematics and reading were less pronounced. Additionally, variations across race/ethnicity groups indicated that students from Group E generally achieved higher reading scores compared to other groups, whereas Group A had comparatively lower performance.

A strong positive relationship was also observed among mathematics, reading, and writing scores, indicating that students who perform well in one subject tend to perform well across others. Overall, the study concludes that both behavioral factors, such as test preparation, and demographic factors, such as gender and race/ethnicity, play a significant role in student academic performance.

Future work could focus on incorporating additional variables such as attendance, study hours, and socio-economic indicators to gain deeper insights into student achievement patterns.

REFERENCES

- [1] Kaggle, "Students Performance in Exams Dataset," Available: <https://www.kaggle.com/datasets/spscientist/students-performance-in-exams/data>
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- [3] G. Siemens and R. S. J. d. Baker, "Learning analytics and educational data mining," *Proceedings of the 2nd International Conference on Learning Analytics and Knowledge - LAK '12*, 2012, doi: <https://doi.org/10.1145/2330601.2330661>.